DRILL 18

RURAL FIREGROUND PRIORITY

NFPA 1001 Level 1 / Rural Operational Standard

CHIMNEY & WOOD STOVE FIRE OPERATIONS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear, SCBA & Fall Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Interior Crew & Roof Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must master standard tactical protocols for rural **Chimney and Wood Stove Fires**—the highest-frequency winter structural call in New Brunswick. Firefighters must perform a TIC scan of the chimney chase and attic for hidden extension, clear burning fuel from the firebox without spreading smoke, deploy chimney flare sticks / dry chemical / drop chains from the roof, avoid thermal shock, and advise on mandatory **WETT inspection** clearance.

📦 TOOLS & APPARATUS SETUP

- Chimney kit: Metal ash bucket, metal shovel/trowel, chimney flare sticks (Chimfex), plastic baggies with dry chemical ("chimney bombs"), weighted drop-chain with wire brush or chains.
- Thermal Imaging Camera (TIC).
- Charged 1.75" backup attack line stretched to the front door (charged safety line).
- Roof ladder and ground ladder for roof access.
- 4-gas detector (monitoring Carbon Monoxide - CO).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE CARDINAL SIN: NEVER spray a solid water hose stream down a hot chimney flue!** Water instantly expands 1,700x into high-pressure steam, causing a steam explosion that shatters clay tile liners, ruptures metal stovepipes, blows burning embers into the living room, and destroys the chimney.

⚠️ **Roof Slip Hazards:** New Brunswick roofs in winter have hidden frost, snow, or ice under chimney soot. Roof ladder deployment is mandatory.

⚠️ **Attic Extension:** A chimney fire burning creosote at >1,000°C (2,000°F) easily conducts radiant heat into adjacent attic wood trusses.

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone approaches an un-secured roof edge or if a heavy water line is pointed toward an open flue.**

PHASE 1: TREAT AS A FULL STRUCTURE FIRE & SIZE-UP (10 MINUTES) — SLOW DRY RUN

1. In rural NB, never treat a chimney call as a "routine service call". Roll with structural apparatus and stretch a charged 1.75" backup line to the front entrance.
2. **Exterior Assessment:** Check chimney cap for heavy rolling sparks, glowing flue pipe, or roaring sound ("freight train").
3. **Interior TIC Sweep:** Crew enters with TIC and immediately scans:
   - Floor 1: Wall area, mantle, and ceiling surrounding the fireplace/stove.
   - Floor 2 (if present): Chimney chase in bedrooms and closets.
   - Attic: Mandatory inspection of where the chimney penetrates the attic floor and roof decking. Look for orange/white heat glow.

DRILL 18: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Fireground Priority • Practical Application

PHASE 2: INTERIOR FIREBOX OPERATIONS (15 MINUTES) — BUDDY COACHING

1. Close wood stove air intake dampers to suffocate combustion airflow.
2. Put on SCBA facepiece (on air) to protect against intense smoke and CO.
3. Open firebox door slowly (avoid sudden oxygen rush).
4. Using metal shovel/trowel, transfer burning logs and coals into a metal ash bucket.
5. Carry the bucket immediately outside to the gravel driveway and douse with water.
6. **Deploying Extinguishing Flare (Chimfex):**
   - Strike the flare cap. Place the burning flare inside the firebox.
   - Close the stove door tightly. The flare consumes available oxygen and releases fire-suppressing chemicals that travel up the flue, knocking out creosote combustion in 60–90 seconds.

PHASE 3: ROOF FLUE EXTINGUISHMENT (20 MINUTES) — TIMED / STRESS

1. **Accessing the Peak:**
   - Raise 24-ft extension ladder to eaves; deploy 14-ft roof ladder over the roof ridge.
   - Firefighter climbs with safety harness and chimney kit.
2. **Method 1: Dry Chemical "Chimney Bombs":**
   - Take 1/2 cup of dry chemical powder sealed inside a sandwich ziplock bag.
   - Drop the bag directly down the chimney flue from the top.
   - The plastic bag melts halfway down the roaring flue, releasing a fine cloud of dry chemical powder that extinguishes the burning creosote instantly without water damage!
3. **Method 2: Weighted Drop Chains:**
   - If heavy creosote or a chimney blockage remains, lower a weighted link chain down the flue.
   - Gently swing and bounce the chain against the inner walls to break loose burning creosote flakes, which drop into the cleared cleanout/firebox below.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Leaving the scene without checking the attic:* Creosote fires often crack the masonry flue inside the attic where nobody looks. The fire smoulders in wood trusses and flashes over 2 hours after the department leaves. Always verify attic clearance with a TIC!
- ❌ *Not red-tagging the appliance:* Inform the homeowner in writing that the heating appliance is **out of service** and cannot be used until professionally inspected and certified by a **WETT (Wood Energy Technology Transfer)** technician.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the TIC reveal thermal conduction through sheetrock?", "Why must we never discharge a fog nozzle down the flue?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Chimney kit restocked: Shovel, metal bucket cleaned, dry chem bags restocked.
- ☐ Ladders returned to truck racks.
- ☐ 4-gas detector powered down and docked in charger.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS